

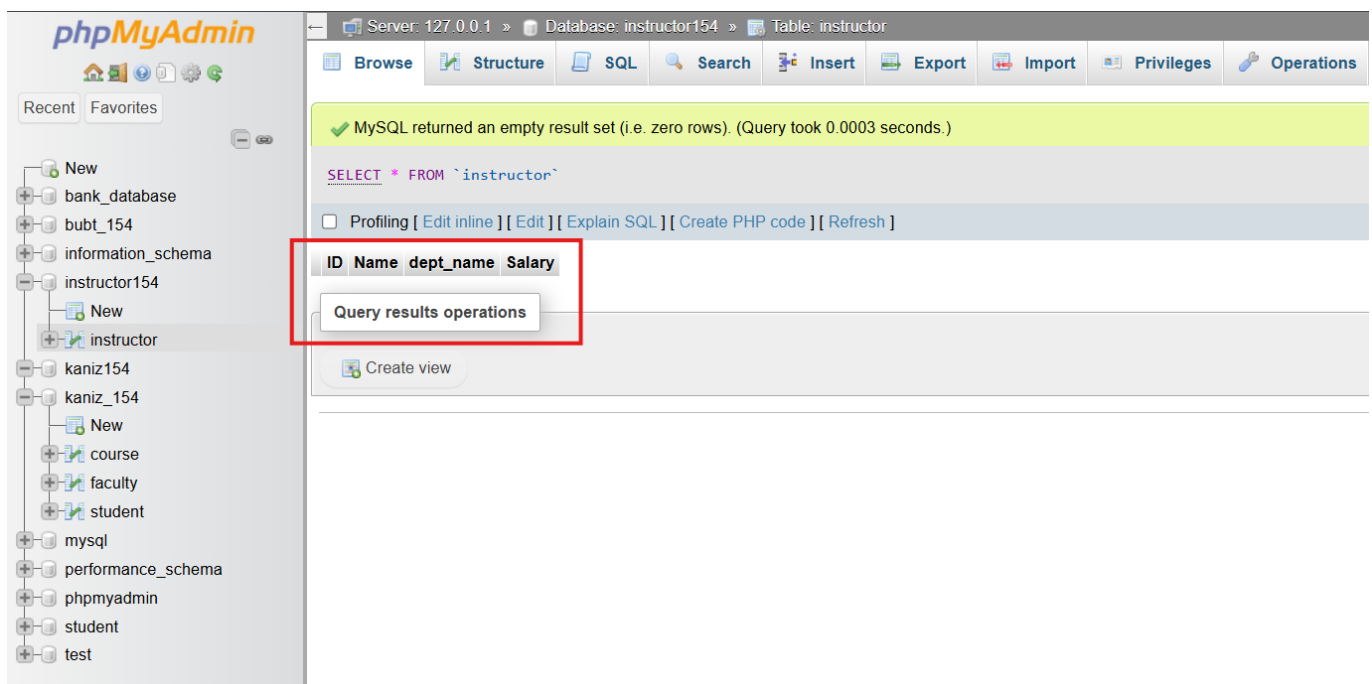
Lab 03: Bank Database

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1. Creating a new database:

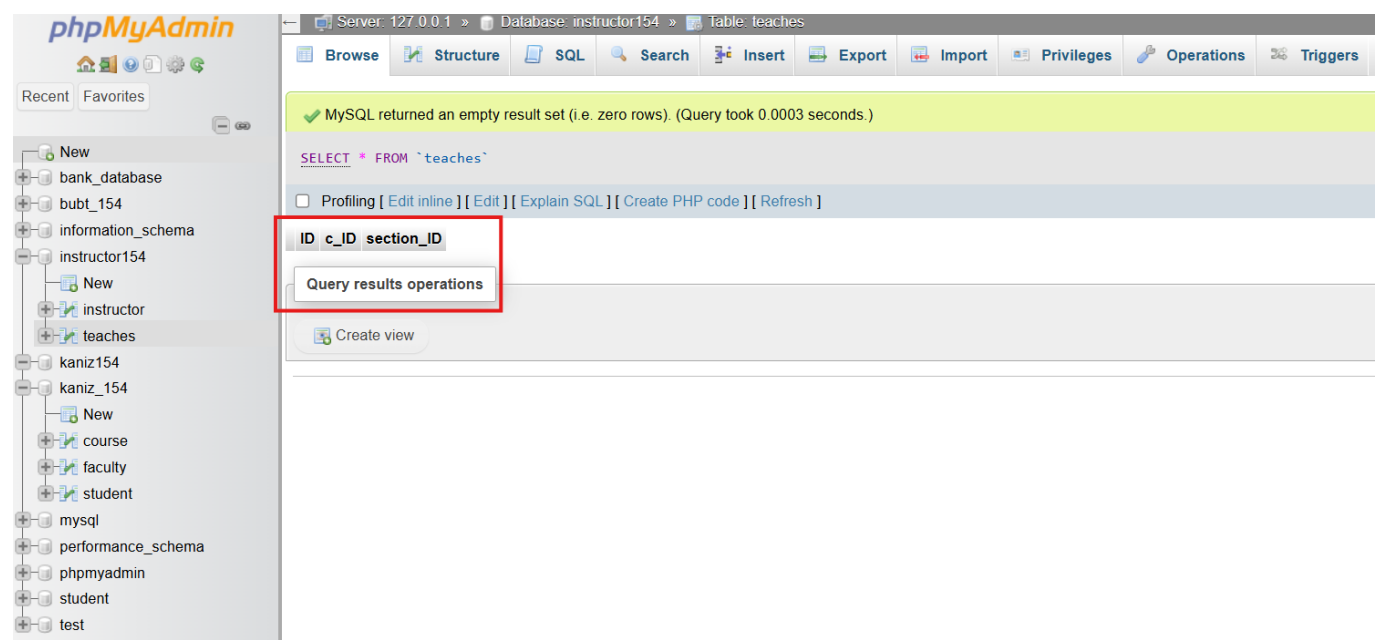
```
CREATE DATABASE bank_database;
```



2. Create 6 tables in this DB named: account, branch, depositor, customer, loan, and borrower

Create Account:

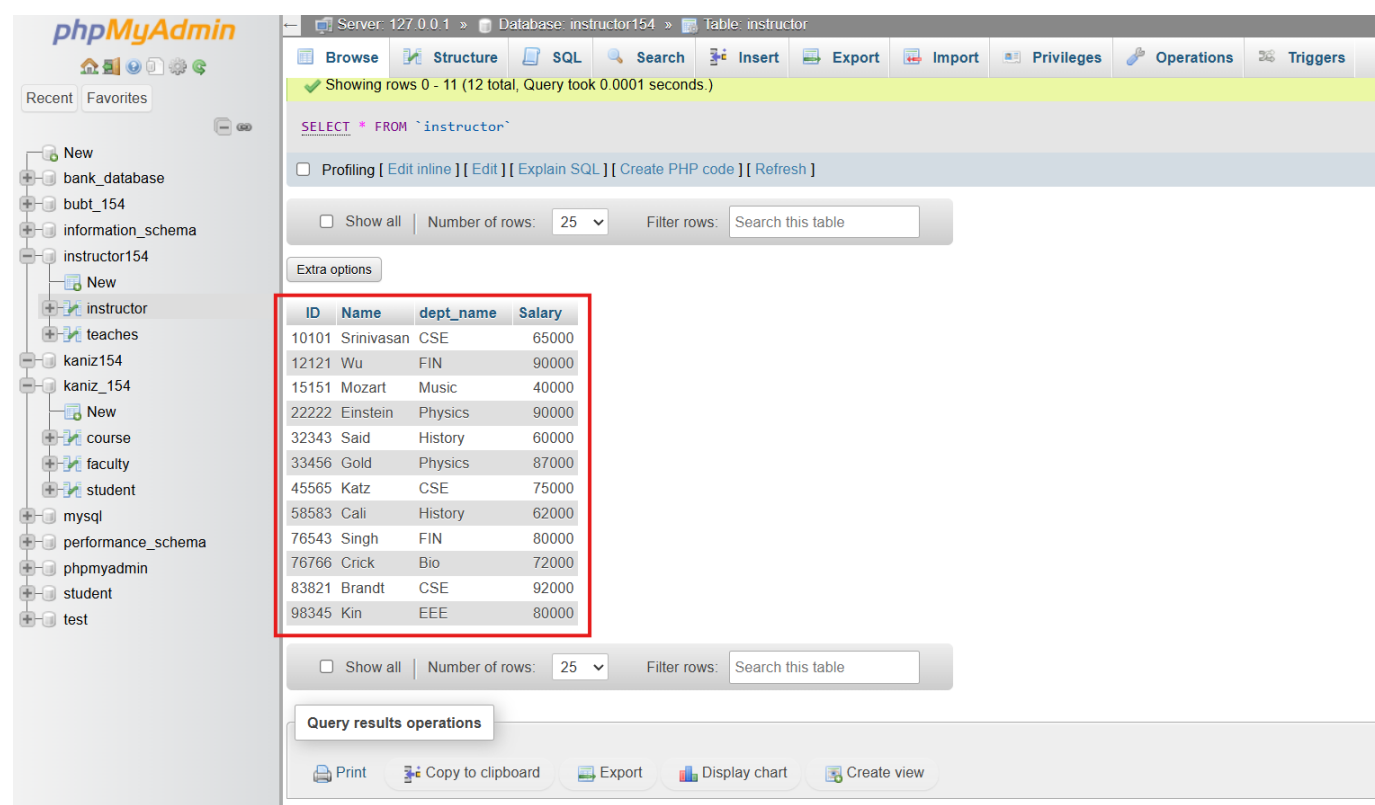
```
CREATE TABLE account (account_number char(5), branch_name varchar(16), balance int(7));  
CREATE TABLE branch (branch_name varchar(16), branch_city varchar(16), assets int(8));  
CREATE TABLE depositor (customer_name varchar(16), account_number char(5));  
CREATE TABLE customer (customer_name varchar(16), customer_street varchar(32), customer_city varchar(16));  
CREATE TABLE loan (loan_number char(5), branch_name varchar(16), amount int(8));  
CREATE TABLE borrower (customer_name varchar(16), loan_number char(8));
```



3. Inserting data into the tables

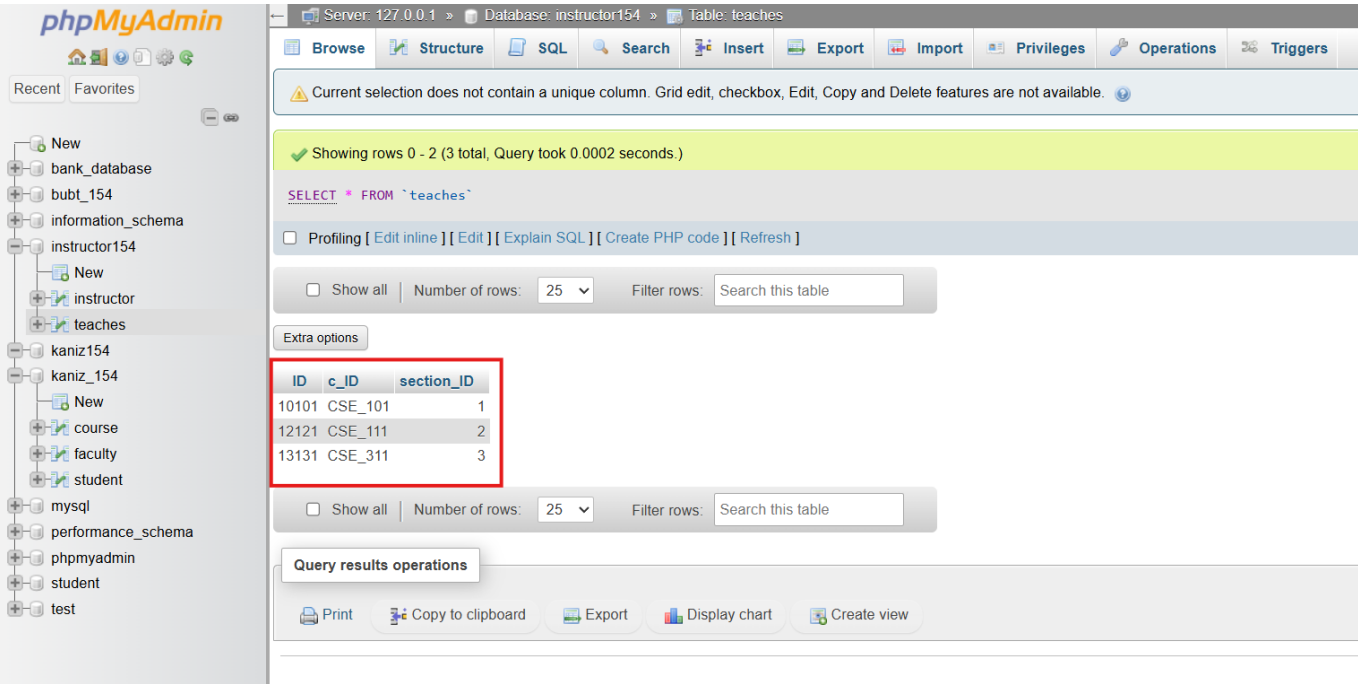
Account:

```
INSERT INTO account VALUES("A-101", "Downtown", 500);
INSERT INTO account VALUES("A-102", "Perryridge", 400);
INSERT INTO account VALUES("A-201", "Brighton", 900);
INSERT INTO account VALUES("A-215", "Mianus", 700);
INSERT INTO account VALUES("A-217", "Brighton", 750);
INSERT INTO account VALUES("A-222", "Redwood", 700);
INSERT INTO account VALUES("A-305", "Round Hill", 350);
```



Branch:

```
INSERT INTO branch VALUES("Brighton", "Brooklyn", 7100000);
INSERT INTO branch VALUES("Downtown", "Brooklyn", 9000000);
INSERT INTO branch VALUES("Mianus", "Horseneck", 4000000);
INSERT INTO branch VALUES("North Town", "Rye", 3700000);
INSERT INTO branch VALUES("Perryridge", "Horseneck", 1700000);
INSERT INTO branch VALUES("Pownal", "Bennington", 3000000);
INSERT INTO branch VALUES("Redwood", "Palo Alto", 2100000);
INSERT INTO branch VALUES("Round Hill", "Horseneck", 8000000);
```



Depositor:

```
INSERT INTO depositor VALUES("Hayes", "A-102");
INSERT INTO depositor VALUES("Johnson", "A-101");
INSERT INTO depositor VALUES("Johnson", "A-201");
INSERT INTO depositor VALUES("Jones", "A-217");
INSERT INTO depositor VALUES("Lindsay", "A-222");
INSERT INTO depositor VALUES("Smith", "A-215");
INSERT INTO depositor VALUES("Turner", "A-305");
```

phpMyAdmin

Server: 127.0.0.1 > Database: instructor154 > Table: teaches

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Show all Number of rows: 25 Filter rows: Search this table

Extra options

ID	Name	dept_name	Salary	ID	c_ID	section_ID
10101	Srinivasan	CSE	65000	10101	CSE_101	1
10101	Srinivasan	CSE	65000	12121	CSE_111	2
10101	Srinivasan	CSE	65000	13131	CSE_311	3
12121	Wu	FIN	90000	10101	CSE_101	1
12121	Wu	FIN	90000	12121	CSE_111	2
12121	Wu	FIN	90000	13131	CSE_311	3
15151	Mozart	Music	40000	10101	CSE_101	1
15151	Mozart	Music	40000	12121	CSE_111	2
15151	Mozart	Music	40000	13131	CSE_311	3
22222	Einstein	Physics	90000	10101	CSE_101	1
22222	Einstein	Physics	90000	12121	CSE_111	2
22222	Einstein	Physics	90000	13131	CSE_311	3
32343	Said	History	60000	10101	CSE_101	1
32343	Said	History	60000	12121	CSE_111	2
32343	Said	History	60000	13131	CSE_311	3
33456	Gold	Physics	87000	10101	CSE_101	1
33456	Gold	Physics	87000	12121	CSE_111	2
33456	Gold	Physics	87000	13131	CSE_311	3
45565	Katz	CSE	75000	10101	CSE_101	1
45565	Katz	CSE	75000	12121	CSE_111	2
45565	Katz	CSE	75000	13131	CSE_311	3
58583	Cali	History	62000	10101	CSE_101	1
58583	Cali	History	62000	12121	CSE_111	2
58583	Cali	History	62000	13131	CSE_311	3
76543	Singh	FIN	80000	10101	CSE_101	1

Show all Number of rows: 25 Filter rows: Search this table

Query results operations

Customer:

```

INSERT INTO customer VALUES("Adams", "Spring", "Pittsfield");
INSERT INTO customer VALUES("Brooks", "Senator", "Brooklyn");
INSERT INTO customer VALUES("Curry", "North", "Rye");
INSERT INTO customer VALUES("Glenn", "Sand Hill", "Woodside");
INSERT INTO customer VALUES("Green", "Walnut", "Stamford");
INSERT INTO customer VALUES("Hayes", "Main", "Harrison");
INSERT INTO customer VALUES("Johnson", "Alma", "Palo Alto");
INSERT INTO customer VALUES("Jones", "Main", "Harrison");
INSERT INTO customer VALUES("Lindsay", "Park", "Pittsfield");
INSERT INTO customer VALUES("Smith", "North", "Rye");
INSERT INTO customer VALUES("Turner", "Putnam", "Stamford");
INSERT INTO customer VALUES("Williams", "Nassau", "Princeton");

```

phpMyAdmin

Recent Favorites

New

bank_database

bubt_154

information_schema

instructor154

New

instructor

teaches

kaniz154

kaniz_154

New

course

faculty

student

mysql

performance_schema

phpmyadmin

student

test

Server: 127.0.0.1 » Database: instructor154 » Table: instructor

Browse

Structure

SQL

Search

Insert

Export

Import

Privileges

Operations

Triggers

Show query box

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 1 (2 total, Query took 0.0002 seconds.)

Select Name, c_ID from instructor INNER JOIN teaches where Instructor.ID=teaches.ID;

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

Name	c_ID
Srinivasan	CSE_101
Wu	CSE_111

Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

Print

Copy to clipboard

Export

Display chart

Create view

Loan:

```
INSERT INTO loan VALUES("L-11", "Round Hill", 900);
INSERT INTO loan VALUES("L-14", "Downtown", 1500);
INSERT INTO loan VALUES("L-15", "Perryridge", 1500);
INSERT INTO loan VALUES("L-16", "Perryridge", 1300);
INSERT INTO loan VALUES("L-17", "Downtown", 1000);
INSERT INTO loan VALUES("L-23", "Redwood", 2000);
INSERT INTO loan VALUES("L-93", "Mianus", 500);
```

phpMyAdmin

Recent Favorites

New

bank_database

bubt_154

information_schema

instructor154

New

instructor

teaches

kaniz154

kaniz_154

New

course

faculty

student

mysql

performance_schema

phpmyadmin

student

test

Server: 127.0.0.1 » Database: instructor154 » Table: instructor

Browse

Structure

SQL

Search

Insert

Export

Import

Privileges

Operations

Triggers

Show query box

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 0 (1 total, Query took 0.0002 seconds.)

Select name, c_ID from instructor INNER JOIN teaches where Instructor.ID=teaches.ID AND dept_name= "CSE";

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

name	c_ID
Srinivasan	CSE_101

Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

Print

Copy to clipboard

Export

Display chart

Create view

Borrower:

```
INSERT INTO borrower VALUES("Adams", "L-16");
INSERT INTO borrower VALUES("Curry", "L-93");
INSERT INTO borrower VALUES("Hayes", "L-15");
INSERT INTO borrower VALUES("Johnson", "L-14");
INSERT INTO borrower VALUES("Jones", "L-17");
INSERT INTO borrower VALUES("Smith", "L-11");
INSERT INTO borrower VALUES("Smith", "L-23");
INSERT INTO borrower VALUES("Williams", "L-17");
```

The screenshot shows the phpMyAdmin interface. On the left is the database navigation tree with the following structure:

- Recent
- Favorites
- New
- bank_database
- bubt_154
- information_schema
- instructor154
 - New
 - instructor
 - teaches
- kaniz154
- kaniz_154
- mysql
- performance_schema
- phpmyadmin
- student
- test

The main panel shows the 'Table: Instructor' view. The top navigation bar includes: Browse, Structure, SQL, Search, Insert, Export, Import, Privileges. Below this is a 'Show query box' button and a warning message: 'Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.'

The query execution results are shown in a green bar: 'Showing rows 0 - 0 (1 total, Query took 0.0001 seconds.)'. The SQL query entered is: `Select count(*) from Instructor;`. Below the query, there are links for Profiling, Edit inline, Edit, Explain SQL, Create PHP code, and Refresh.

The results section shows a single row with the value '12' for the `count(*)` column. The 'Extra options' button is visible. Below the results, there are controls for 'Show all', 'Number of rows' (set to 25), and 'Filter rows' (Search this table).

The 'Query results operations' bar at the bottom includes buttons for Print, Copy to clipboard, Export, Display chart, and Create view.